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5 / 5 submitted

Q1. Harshad number

ANSWER

```
def digit_sum(n: int) -> int:
    s = 0
    while n > 0:
        s += n % 10
        n //= 10
    return s

# Read input number
N = int(input("Enter a number: "))

# Check if N is a Harshad number
if N > 0:
    ds = digit_sum(N)
    if ds != 0 and N % ds == 0:
        print(f"{N} is a Harshad number.")
    else:
        print(f"{N} is not a Harshad number.")
else:
    print("Harshad number check is for positive integers only.")

# Print all Harshad numbers between 1 and 500
print("Harshad numbers between 1 and 500:")
for x in range(1, 501):
    ds = digit_sum(x)
    if ds != 0 and x % ds == 0:
        print(x, end=" ")
print()
```

Q2. GCD

CODE

```
a = int(input("Enter first: "))
b = int(input("Enter second: "))
while b != 0:
    temp = a
    a = b
    b = temp % a
print("GCD is", a)
```

INPUT

36
60

OUTPUT

Enter first: Enter second: GCD is 12

Q3. sum to a target value

ANSWER

```
def find_pairs(lst, target):
    seen = set()
    pairs = []
    for num in lst:
        complement = target - num
        if complement in seen:
            pairs.append((complement, num))
        seen.add(num)
    return pairs
data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = find_pairs(data, t)
print("Pairs:", result)
```

Q4. Bitwise Operators

ANSWER

```
a=13, b=21, c=26, result=31  
0b111111 0o37 0x1f  
7 62 -32
```

Q5. binary search

CODE

```
def rotated_binary_search(arr, target):  
    low = 0  
    high = len(arr) - 1  
  
    while low ≤ high:  
        mid = (low + high) // 2  
  
        if arr[mid] == target:  
            return mid  
  
        if arr[low] ≤ arr[mid]:  
            if target ≥ arr[low] and target < arr[mid]:  
                high = mid - 1  
            else:  
                low = mid + 1  
        else:  
            if target > arr[mid] and target ≤ arr[high]:  
                low = mid + 1  
            else:  
                high = mid - 1  
  
    return -1
```

```
return -1
```

```
data = [4, 5, 6, 7, 0, 1, 2]  
t = int(input("Target: "))  
result = rotated_binary_search(data, t)  
print("Found at:", result)
```

OUTPUT

Target: Found at: 4